Deterioration states — methodology

Collections risk modelling · deterioration states · docs/deterioration_states/METHODOLOGY.md

How a credit-card account in buckets 4–7 fell into delinquency, and what it is doing now, read from its DPD bucket history only. One state per account, assigned by a short decision tree. MECE by construction, simple enough to explain to the risk team in one slide, and designed to stay out of the way of the other persona axes.

> **Status.** Designed and validated on calibrated **synthetic** data. Not yet run on the bank's book. Every number below marked *(synthetic)* is from simulation and is a *method check*, not a finding about the portfolio. The real-data run is `snippets/dpd_states_validate.py`; its expectations are in §8.

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1. Where this sits among the persona axes

Axis	Question it answers	Inputs
Locatability	Can we reach them?	Reachability flag, in-country flag
Ability	Can they pay?	Score: salary, inflow, CASA balance
Payment credibility	Do they pay and keep their word?	Score: promises kept (0–2), money coming in last 3 months (0–2), pre-delinquency history clean (0–1)
Deterioration (this)	**How did they fall in, and are they stabilized now?**	**DPD bucket history only**

Design rules agreed with the business owner:

1. **MECE** — every account gets exactly one state; no "other" bucket.
2. **Simplicity beats precision** — a rule a collector can repeat beats a better-fitting one nobody can explain. Thresholds are round numbers.
3. **Orthogonal where possible** — the states must not rebuild the credibility or ability score. Known overlaps are named (§7), not hidden.
4. **5–6 states, at most 8**, each with a different action for the risk team.

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2. Population and data conventions

- **Spine:** credit-card accounts whose **current** bucket is 4, 5, 6 or 7. Bucket 8+ is a separate category. Full history is available for every account.
- **The string:** one digit per month, **leftmost = current month** (`54320002000` → today 5, last month 4, …). Configurable if a source is reversed.
- **0 ≡ 1.** Bucket 1 (a few days late) is treated as normal.
- **Monthly movement `d` = bucket this month − bucket last month**, which reads as a payment:

d	Meaning	Months paid that month (= 1 − d)
+1	roll — paid nothing (or less than one month's due)	0
0 at bucket ≥ 2	**hold** — paid one month's due, stays in the same bucket	1

d	Meaning	Months paid that month (= 1 − d)
−k	paid k + 1 months, moved back k buckets	k + 1
back to 0/1	**cure** — common from 2–3, very rare from 4+	—

- **Spell:** a continuous run of months at bucket ≥ 2. The *current spell* is the one the account is in today.
- **Blip:** an earlier spell of 1–2 months that never passed bucket 2 (`0 0 2 0 0`). Blips do not count as earlier spells and do not end the clean history.
- **Billing-cycle jumps:** a +2 move next to a hold (`3 3 5` or `3 5 5`) is the reporting artefact of a straight roll and is repaired to `3 4 5` . Any other +2 is read as a roll.

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3. Features used by the states

Only features that **do not drift with time since reaching 4** are used to assign states. For most of the spine the current bucket is a clock (one bucket per month), so any count over a fixed window, or any rate over the whole spell, changes with the bucket rather than with behaviour.

Feature	Definition	Fixed once past 4?
`months_before_spell`	months of history before the current spell started	yes
`n_blips`	earlier excursions of 1–2 months that never passed bucket 2 (EPD allows one)	yes
`n_prior_episodes`	earlier real (non-blip) spells (EPD requires none)	yes
`max_prior_bucket`	deepest bucket reached in any earlier (non-blip) spell; 0 if none (repeat delinquent)	yes
`runup_len`	months from the last clean month to the first month at bucket 4 (3 = straight 2→3→4)	yes
`runup_pays`	months on the way to 4 in which 2+ months were paid (moved back)	yes
`last6_in_band`	of the last 6 months (incl. today), how many were within today's bucket ±1	no — current behaviour, by design

Other features (`runup_holds` , `post4_paid_rate` , `max_hold_run` , …) are computed for diagnostics and discovery only.

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4. The states

Questions are asked **in order**; the first "yes" assigns the state.

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Q1  Early payment default?  <= 3 months of history before this spell
                            (at most one blip inside it — a payoff from 2 — and no
                            earlier real spell) AND run-up to 4 took <= 3 months
                                                    -> early_payment_default
Q2  Stabilized?             >= 5 of the last 6 months within today's bucket ±1
                                                    -> stabilized
Q4  Repeat delinquent?      [B, C] an earlier spell reached bucket 3 and cured
                                                    -> repeat_delinquent
Q5  Straight roller?        run-up (last clean month -> first month at 4) <= 3 months
      yes -> straight_roller  [B, C: >= 12 months of history before the spell
                               -> straight_roller_seasoned, else
straight_roller_early_life]
      no  -> slow_roller      [C: paid 2+ months at once on the way down
                               -> slow_roller_erratic, else slow_roller_intermittent]
```

(Q3 "paying since bucket 4" was tested and rejected — §6.4.)

Three options

Option	States	Use when
A — minimal (4)	EPD, stabilized, straight roller, slow roller	Strict orthogonality: prior-spell history is left entirely to credibility
B — balanced (6), recommended	EPD, stabilized, repeat delinquent, seasoned straight roller, early-life straight roller, slow roller	Default. Matches the natural groups in the data (§6.1) and the 5–6 target
C — detailed (7)	B with slow roller split into intermittent payer / erratic payer	The risk team has a different action for each and both are ≥ 4%

What each state means and the action it points to

State (display name · `code`)	Reads as	Action flavour	Share *(synthetic)*
Early payment default (EPD) · `early_payment_default`	At most 3 months of history (one blip allowed), rolled straight in, never really paid	Early-risk / first-party-fraud review, early legal; no restructuring	6% (bank: 6%)
Stabilized · `stabilized`	Paying about one month's due each month for 5 of 6 months	**Offer restructuring to break the hardship**	7% (bank: 6%)
Repeat delinquent (re-default) · `repeat_delinquent`	Has been here before and cured from bucket 3	Has shown they can catch up: firm promise-based plan; credibility decides the pressure	16%
Seasoned straight roller · `straight_roller_seasoned`	At least a year on book (≥ 12 months clean), then rolled every month without paying	Life event. Read with the other axes: not in country → departed; no salary → income shock; salary still in → won't pay. Most likely to respond to a hardship route	52%

State (display name · `code`)	Reads as	Action flavour	Share *(synthetic)*
Early-life straight roller · `straight_roller_early_life`	4–11 months on book, then rolled every month without paying	Same triage; also an underwriting signal for acquisition	3% — below the 5% floor in synthetic data; the real share decides (E16)
Slow roller (partial payer) · `slow_roller`	Paid something on the way down but kept rolling	Over-indebted: longer-tenor restructure or settlement	16%
↳ **Intermittent payer** *(C)* · `slow_roller_intermittent`	Paid one month here and there, never caught up	Chronic insufficiency: restructure to what they actually pay	10%
↳ **Erratic payer** *(C)* · `slow_roller_erratic`	Caught up 2+ months at once, then rolled again	Lumpy / irregular income: plan aligned to inflow timing	6%

In option A the two straight-roller states are one: **Straight roller** · `straight_roller`.

Stabilized is the only state that moves month to month — it describes what the customer is doing now (*synthetic:* 62% are still stabilized next month). Every other state is fixed once the account is past bucket 4 (*synthetic:* 97–100% keep their state next month).

4.1 Names

The names use established card-risk and collections terms, and follow two rules:

1. **Describe behaviour, not intent.** These labels reach collector screens, audit trails and possibly complaint files. They say what the DPD history shows ("never really paid", "rolled every month", "paid one month at a time"), never why ("fraud", "won't pay"), which the history cannot prove.
2. **Code names are stable; display names are free.** Code uses short identifiers (`early_payment_default`, `straight_roller_seasoned`, …). `DISPLAY_NAMES` and `REMARKS` in `dpd_states/states.py` carry the business labels, so renaming for the risk team never breaks the pipeline or old results. Every report starts with the names table.

Term	Where it comes from	Note
Early payment default (EPD)	Early-risk monitoring (FPD / SPD / TPD = first / second / third payment default; EPD = default early in the life of the account)	**Formerly "bust-out"** — the bank's internal name, kept in remarks. Classic *bust-out* fraud is different: a customer builds a good record and limit, maxes out (often with cash advances) and disappears. Its DPD history is clean-then-straight-roll, so it lands in **seasoned straight roller**; spotting it needs spend / utilisation spikes before delinquency — a fraud overlay, not this axis
Stabilized	The bank's own term; roll-rate "stay" behaviour	Industry has no single standard label for "pays enough to stay in the bucket"
Repeat delinquent / re-default	Collections recidivism; re-default rates after cure or restructure	
Straight roller	Core card-collections term: rolls through every bucket without a payment	

Term	Where it comes from	Note
Seasoned / early-life	Seasoning and months on book (MOB); the first 12 months is the standard early-life window	
Slow roller (partial payer)	Roll-rate language: the counterpart of the straight roller	**Also called partial payer** — the more common collections term. The rejected Q3 state was renamed `paying_since_4` so "partial payer" means only this state
Intermittent / erratic payer	Collections payment-pattern labels	

5. Thresholds and the evidence behind them *(synthetic sweeps)*

Threshold (`parameter`)	Value	Why this value	What moving it does
EPD history (`epd_max_history`, `epd_max_blips`)	**≤ 3 months, at most one blip**	Business definition (agreed): a customer who never pays misses the first or second bill, so 2–3 clean months at most (`0 0 2 3 4`, `0 0 0 2 3 4`, `0 2 0 2 3 4`). No natural knee in the data	*Synthetic:* at 3 months 93% of the state is a planted early defaulter; at 6 months 79%, at 12 months 55% — longer histories are early-life defaults and land in early-life straight roller. Sweep reported in every run (2–12 months)
EPD earlier spells	**none; one blip allowed**	An early defaulter may pay off once, but only from bucket 2 (a blip)	`0 2 3 4 0 0 2 3 4 5 6` is therefore repeat delinquent, not EPD; two blips (`0 2 0 2 0 2 3 4`) is not EPD
Straight-roller run-up (`straight_max_runup`)	**≤ 3 months**	The one sharp threshold: any single paid month on the way down makes it a slow roller	≤ 4 moves ~70% of genuine slow rollers into straight roller
Stabilized band (`stabilized_min_in_band`)	**5 of last 6 months within ±1**	The bank's existing stabilized rule, unchanged	4 of 6: share ~14%, next-month roll ~46%, persistence ~63%. 6 of 6: ~5%, ~32%, ~80%
Repeat-delinquent depth (`repeat_min_prior_peak`)	**earlier spell reached 3**	The bank counts ~85% single-spell; spells from 3 give 83% *(synthetic)*	From 2: 19%. From 4: 5% (cures from 4+ are rare)
Seasoned (`seasoned_min_months`)	**≥ 12 months**	Business definition (agreed): the first year on book is the standard early-life window. EPD already takes ≤ 3 months, so early life = 4–11 months	*Synthetic:* 12 m gives early life only 3.4% (below the 5% floor), 18 m 7%, 24 m 11%, 36 m 20% — but these shares come from the generator's assumed tenure distribution, so the real share decides (E16)

Rule order. EPD is asked before stabilized (the bank's order). Only ~0.2% of the spine is an early defaulter that is also stabilized today, so the order barely matters.

6. What was tried and dropped — and why

6.1 Unsupervised clustering as the source of states (k-means, GMM)

- **Whole spine, k-means:** very stable (bootstrap ARI 0.99) but only ARI 0.3–0.4 against the planted patterns and 0.1–0.3 against the rule states. It always merged holders with staircases and never isolated slow slides. **Stability does not mean validity** — on real data stability is the only check available, and it cannot detect merged behaviours.
- **GMM:** BIC never reached a minimum (kept improving to 12 components) because the features are spiky — 56–93% of accounts sit on a single value (run-up = 3, no earlier spell, …), and GMM fits narrow components to the spikes. Under 1% of accounts were ambiguous, so the soft-assignment advantage disappears. Best ARI vs planted 0.48 but stability fell to 0.67–0.93. Not used. GMM fits continuous inputs (ability scores, balances) far better than DPD-derived counts.
- **Residual clustering (after EPD and stabilized) — kept as a diagnostic.** k-means at k = 6 on the remaining ~88% found six clean, nameable groups: long-clean straight roll, short-clean straight roll, earlier spell to 3+, paying after 4 but still rolling, held on the way, paid 2+ on the way. With the final thresholds, agreement at k = 6 is A 0.11, **B 0.32, C 0.34** (best at k = 3: B 0.65 vs A 0.31) — the evidence that two residual states were too few. Every B/C split maps to one of these clusters with a one-line rule; the clusters themselves are not the states. The clusters split clean history near 36 months, but that reflects the generator's assumed tenure distribution; the seasoned cut is a business definition (12 months, §5).

6.2 A "mixed" / residual state

Dropped. An ordered tree with a default branch is MECE without one, and a residual bucket invites "everything we don't understand" to grow.

6.3 Staircase (holds for months at each level, then steps up)

Merged. While it is holding a bucket it *is* stabilized (that is the actionable fact); just after a step it reads as a slow roller. Telling a genuine holder from a staircase mid-hold needs more than 6 months of history and an extra rule, and the next-month risk was the same (*synthetic:* 0.28 vs 0.29). It separates only over 6 months (exit 33% vs 50%).

6.4 "Paying since bucket 4" — pays after reaching 4 but still rolling (Q3)

Tested at 0.34 / 0.5 / 0.67 months paid per month since 4. Rejected:

- barely separates risk (next-month roll ~80% vs ~86%);
- unstable — ~37% leave it every month — and it **drags the stability of the fixed states from ~98% to ~91%**, because accounts flip in and out of it;
- 79% of it is "paid in the last 3 months" — the credibility axis's 2-point component.

It is one of the six natural residual clusters, so it is real behaviour — but it belongs to credibility, not here. The switch is kept in code (`paying_since_4_min_rate` , state `paying_since_4` — renamed from "partial payer" so that term means only the slow roller) so the test can be repeated (expectation E24).

6.5 Separate states for slow slide, dip-and-roll, volatile

Merged into slow roller (A, B) or split once by "paid 2+ months on the way" (C). The three confuse each other in every view (shapes, clusters, trees), and the risk actions are close. Much of "volatile" (cure then relapse) is a repeat delinquent by definition.

6.6 Repeat delinquent in option A

Left out of A on purpose. Repeat delinquent is built from earlier spells — the same history credibility's 1-point "pre-delinquency clean" flag scores. In the self-test (with a clean point defined as "no earlier spell") the states explained **84%** of that point's variance. It is included in B/C because it is only 1 of credibility's 5 points and it carries more than a yes/no: *cured from bucket 3 before* is evidence of catch-up capacity. Expectation E19b measures the real overlap and decides A vs B.

6.7 "Stabilized or improving" clause (today's bucket ≤ 6 months ago)

Dropped. It changed **0.0%** of assignments. Straight / slow roller describe how the account fell in, so an improving account carries no misleading label without it.

6.8 Payment-based refinement of stabilized

Tested: "≤ 1 roll in 6 months AND paid 2+ months at least once" separated holders from staircases at 99% precision — but only because the synthetic generator built it that way, and it leans on payment behaviour (credibility). Replaced by the bank's own 5-of-6 rule.

6.9 Features that drift with time since bucket 4

Dropped from assignment: whole-spell roll rate, fixed-window counts (rolls in the last 6 or 12 months), months paid since 4 as a count. Straight rollers at bucket 4 and 7 looked different only because of the clock. Replaced by run-up features (fixed once past 4) and rates within the current spell.

6.10 Shape table as the assignment

Kept for discovery and stakeholder examples only. ~5–6k distinct banded shapes; the top 30 cover ~35–40% of accounts — too fragmented to assign everyone.

6.11 n-gram ranking by planted-label purity

Fixed. The first version had a meaningless "lift" (per-account presence divided by per-month code frequencies, rescaled by an arbitrary constant) and ranked by planted labels that real data does not have. Lift is now measured against each account's own codes in shuffled order: lift > 1 means the order of moves matters.

6.12 Treating +2 jumps as impossible

Replaced by repair (+2 next to a hold → straight roll) and "otherwise a roll". Real data has billing-cycle effects (example: `…3 4 3 5 4…`).

6.13 Payment-promise or inflow inputs in any state

Never used — credibility and ability own them. Their overlap is *measured* (§7).

6.14 EPD ("bust-out") with up to 6 months of history

The first version allowed ≤ 6 months before the spell, chosen to match "a few months". It made `0 0 0 0 0 0 2 3 4` a bust-out (now EPD) — but six clean months means about four bills were paid (or nothing was due): an **early-life default**, not a customer who never paid. The synthetic 6% match did not support 6 months either: the generator's bust-out volume had been tuned to hit 6% *under that rule*, so the match was circular. Replaced by the agreed business definition: **≤ 3 months, at most one blip**. Under it the generator still gives 5.8% EPD; every planted early defaulter is caught, and 93% of the state is a planted early defaulter (79% under the old 6-month rule).

6.15 Seasoned / early-life split in option A

Not in A because, in synthetic data, it adds no risk separation (both ~94–95% exit within 6 months, by construction). It is in B/C for its action difference (life event vs underwriting signal). Real outcomes decide whether it earns its place.

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7. Overlap with the other axes (known and measured)

Overlap	Size *(synthetic)*	Treatment
stabilized ↔ credibility "money coming in" (2 pts)	100% of stabilized have money in; 44% of the rest do too	One-directional: stabilized is stricter (enough to hold the bucket, 6 months). Accept and state it. E21
repeat delinquent ↔ credibility "pre-delinquency clean" (1 pt)	eta² 0.84 of the clean point explained by B states	Decides A vs B. E19b
straight / slow roller ↔ ability	none in synthetic (by construction)	Straight rolling may proxy income loss on real data — fine, but say so. E20
any state ↔ locatability	none expected	E19/E20 report it

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8. Real-data validation

How the backtest works from a single table

Today's string holds the past: dropping its newest *k* digits gives the string as it was *k* months ago, and the dropped digits are what happened next. So one table of latest-month strings gives the spine as of *k* months ago **and** its outcomes (next-month roll, exit to 8+ within *k*, cure, next-month state). This needs the table to **keep accounts that have since rolled to 8+ or closed** — a survivor-only table biases the outcomes (the snippet detects and warns).

Expectations (checked automatically — `dpd_states/expectations.py`)

	Check	Expected	Basis
E01	Exactly one state per account	True (FAIL otherwise)	Tree with default branch
E02	Unparseable strings	≤ 2% (FAIL)	CHAR_MAP / STRIP_CHARS set right
E03	Backtest monthly roll rates vs 70/80/90/95%	≤ 7pp off (FAIL)	Confirms orientation and reconstruction
E04	Histories starting inside the spell	≤ 10%	Full history exists
E05	Unrepaired +2 jumps	≤ 2%	Most jumps sit next to a hold
E06	Bucket mix declines 4 → 7	yes	Rising roll rates
E07	One spell only (from bucket 3)	78–92%	Bank ~85%
E08 / E09	EPD / stabilized share	4–8% / 4–9%	Bank 6% / 6%
E10 / E11	Recall of the bank's existing flags	≥ 80% / ≥ 90%	Same definitions

	Check	Expected	Basis
E12	Stabilized next-month roll below the fixed states	≥ 20pp gap	*Synthetic:* 40% vs ~85%
E13	Fixed states keep their state next month	≥ 95%	Run-up fixed once past 4
E14	Stabilized persistence next month	45–85%	Current behaviour
E15	Option A straight-roller share	55–80%	Most roll straight
E16	Smallest option-B state	≥ 5%	Actionable size
E17	Smaller slow-roller split (C)	≥ 4%	Actionable size
E18	Stabilized at bucket 4 vs 7	≥ 2×	90–95% roll at 6–7
E19 / E19b / E20	eta² of credibility / clean point / ability by state	≤ 0.25 / ≤ 0.50 / ≤ 0.10	Orthogonality
E21	Stabilized with money coming in	≥ 90%	Stabilized = paying
E22	Residual clusters favour B over A	ARI gain ≥ 0.15	*Synthetic:* 0.20
E23	One paid month matters (slow roller at run-up ≤ 4 / ≤ 3)	≤ 0.70	Most slow rollers held once
E24	"Paying since 4" stays rejected	persistence ≤ 75%	*Synthetic:* 63%
E25	INFO: exit spread across fixed states	report	Unknowable from synthetic

Every check carries an "if not met" action in the code and in the report.

Choosing the option on real data

1. Any **FAIL** → fix config/data first; read nothing else.
2. **E19b high** (repeat delinquent ≈ credibility clean point) **and** the risk team does not want the cured-before nuance → **A**. Otherwise **B**.
3. **E17 passes and** the risk team has distinct actions for intermittent vs erratic payers → **C**.
4. **E25 ≥ 5pp** → the mechanism carries risk on its own; report it to the risk team as a bonus.

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9. Validation done so far *(synthetic)*

- **Calibrated generator** (`dpd_states/synth.py`): eight planted journey patterns (bust-out, clean abrupt shock, slow slide, staircase, stabilized, repeat episodes, dip-and-roll, volatile); full lifecycles to exit; length-biased snapshot like a real spine; payment holidays (frozen buckets) and billing-cycle +2 jumps; one continuous tenure distribution. Matches the bank's figures: roll rates 70/80/89/95% (target 70/80/90/95%), EPD 5.8% (6%), stabilized 6.5% (6%), single spell 83% from bucket 3 (85%).
- **Method report:** `validation/dpd_states_validation_*.md` (`python -m dpd_states.run_synthetic`).
- **Real-data path self-test:** `validation/selftest_report_*.md` (`python -m dpd_states.selftest`) runs the exact analysis and expectations the workplace run uses, on synthetic data with fake axis columns. All checks pass except two expected WARNs: E19b (repeat delinquent ≈ credibility clean point, §6.6) and E16 (early-life straight roller 3.4% under the 12-month seasoned cut — a property of the synthetic tenure distribution, §5).

What synthetic data cannot tell you: whether the rolling states differ in outcome (they share post-4 dynamics by construction), how big the real overlaps with credibility and ability are, and how much payment

holidays contaminate stabilized. The real-data run answers all three.

10. Open decisions

1. Option A, B or C — decide from E19b, E17 and the risk team's actions (§8).
2. Spell definition (from bucket 3 or 4) — E07 shows which matches the bank's 85%.
3. EPD history threshold — confirm 3 months (one blip) against the bank's existing bust-out rule (E10).
4. Payment holidays / restructurings — if flags exist, exclude frozen months before Q2.

Glossary

Bucket — DPD band, one digit per month · Roll — +1 bucket, nothing paid · Hold — same bucket, one month's due paid · Cure — back to 0/1 · Spell — continuous months at ≥ 2 · Blip — 1–2 month spell never past 2 · Run-up — last clean month to first month at 4 · Spine — accounts currently at 4–7 · eta^2 — share of an axis score's variance explained by the states (0 = orthogonal) · ARI — agreement between two groupings (0 = chance, 1 = identical).